



KTU NOTES

The learning companion.

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NOTIFICATIONS | SOLVED QUESTION PAPERS**

Website: www.ktunotes.in

03/11/2022

MODULE-4

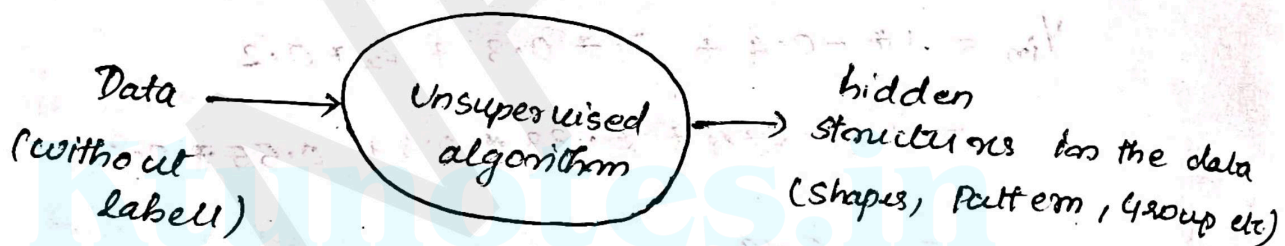
(Unsupervised learning - clustering) + PCA.

- K-means algorithm.
- Hierarchical clustering
- Expectation maximization (EM)

↓
Dimensionality reduction.

Unsupervised learning.

It is a technique used to find hidden structures in the i/p data.



In the case of an un-supervised learning there ~~is~~ ^{is} no training phase so no training data is needed.

Applications of Unsupervised learning

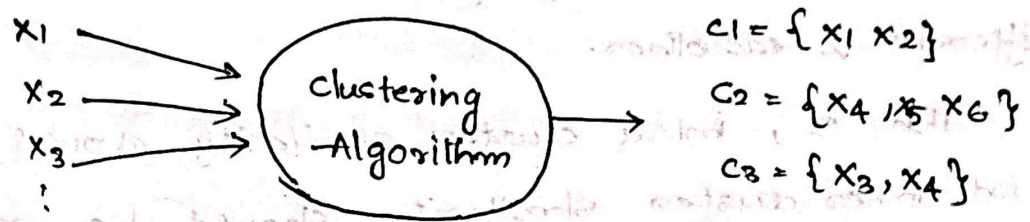
Usually used for clustering related application

Clustering! The task of grouping set of objects. In such a way that objects in the same group (cluster) are more similar to each other than the objects in other clusters

Major clustering approaches

- ① Partitioning based clustering (Kmeans, K-medoids)
- ② Hierarchical clustering (Top-down, Bottom up)
- ③ Density based clustering (DBSCAN)

④ Graph based



- * Instances belonging to one cluster should show similar characteristics to each other. This is called intra cluster similarity.
- * Instances belonging to different clusters should show dissimilar characteristics this is called inter-cluster similarity.
- * The Quality of the clustering algorithm is measured using inter cluster similarity and intra cluster similarity.

K-means algorithm

i/p $\Rightarrow$ Given, m objects

$$X = \{x_1, x_2, \dots, x_m\}$$

g features of objects.

$$x_i = \{x_{i1}, x_{i2}, \dots\}$$

O/p : k clusters

$$S = \{s_1, s_2, s_3, \dots, s_k\}$$

That is,

we have to find k clusters that should satisfy below optimization criteria.

objects within the clusters should be similar to each other and the objects belonging to different clusters should be different to each other.

That is, intra cluster similarity should be maximum and inter cluster similarity should be minimum.

Method:

Given K

- Step 1: Randomly choose K data points [seeds] to be the initial cluster centre.
- Step 2: Compute the distance from each data point to the cluster centers. and ^{assign} each data point to the nearest cluster centre.
- Step 3: Re-compute cluster centers. using current cluster members. Then these cluster centers are taken as new-seeds.
- Step 4: Then check the stopping condition.

If stopping condition is not met then go to step-2

STOPPING CONDITION

1. No change of centroid (cluster centre)
2. No re-assignment of data points
3. Minimum decrease in Sum of Squared Error (SSE)

$$SSE = \sum_{i=1}^K \|x_i - \mu_i\|^2$$

04/11/2022

Perform K-means on below data.

x_1	1	2	2	3	4	5
x_2	1	1	3	2	5	3

 $\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4, \vec{x}_5, \vec{x}_6$

let $\vec{v}_1 = (2, 1)$ & $\vec{v}_2 = (2, 3)$ are the initial cluster centers (seed)

iteration 0: $(2, 1)$ $(2, 3)$

(i) data point	distance from $\vec{v}_1$	distance from $\vec{v}_2$	minimum distances	Assignment cluster
$x_1 (1, 1)$	$\sqrt{x_1^2 + x_2^2} = \sqrt{1+0} = 1$	2.24	1	$\vec{v}_1$
$x_2 (2, 1)$	0	$\sqrt{0} = 0$	0	$\vec{v}_1$
x_3	0	0	0	$\vec{v}_2$
x_4	1.41	1.41	1.41	$\vec{v}_1$
x_5	2.82	2	2	$\vec{v}_2$
x_6	5	3.61	3.61	$\vec{v}_2$

$$* C_1 = \{ \vec{x}_1, \vec{x}_2, \vec{x}_4 \} \rightarrow \vec{x}_2 (\vec{v}_1)$$

$$C_2 = \{ \vec{x}_3, \vec{x}_5, \vec{x}_6 \} \rightarrow \vec{x}_3 (\vec{v}_2)$$

Recalculate cluster centers of C_1 & C_2 .

$$\vec{v}_1 = \frac{1}{3} (1+2+3, 1+1+0)$$

$$= \frac{1}{3} (6, 4) = (2, 1.33)$$

$$\vec{v}_2 = \frac{1}{3} (2+4+5, 3+3+3)$$

$$= \frac{1}{3} (11, 9) = (3.67, 3)$$

→

Iteration 0

New cluster centers are not same as the old cluster centers.

So go to the next iteration, with new cluster centers

Iteration 1

$(2, 1.33)$ $(3.69, 3)$

data point	distance $\vec{v}_1$	distance $\vec{v}_2$	minimum distance	Assigned cluster.
$\vec{x}_1 (1,1)$	1.05	3.77	1.05	$\vec{v}_1$
$\vec{x}_2 (2,1)$	0.33	3.14	0.3	$\vec{v}_1$
$\vec{x}_3 (2,3)$	1.67	1.80	1.67	$\vec{v}_1$
$\vec{x}_4 (3,2)$	1.20	1.80	1.20	$\vec{v}_1$
$\vec{x}_5 (4,3)$	2.6	0.75	0.72	$\vec{v}_2$
$\vec{x}_6 (5,3)$	4.4	1.89	1.89	$\vec{v}_2$

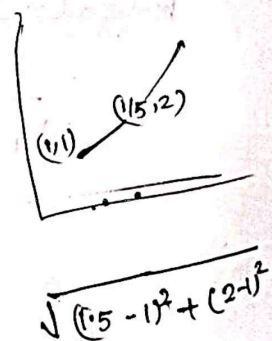
$$C_1 = \{ \vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4 \}$$

$$C_2 = \{ \vec{x}_5, \vec{x}_6 \}$$

Recalculate cluster centers of C_1 & C_2 .

$$\begin{aligned} \vec{v}_1 &= \frac{1}{4} (1+2+2+3, 1+1+3+2) \\ &= \frac{1}{4} (8, 7) = (2, 7/4) \end{aligned}$$

$$\begin{aligned} \vec{v}_2 &= \frac{1}{2} (4+5, 3+3) \\ &= \frac{1}{2} (9, 6) \\ &= (9/2, 3) \end{aligned}$$



8/11/2022

Tutorial hr

using k-means algorithm find two clusters in the following data

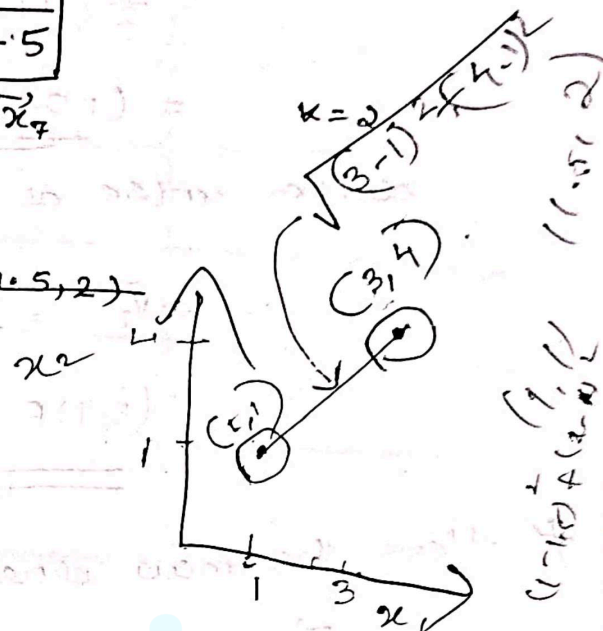
x_1	1	1.5	3	5	3	4	3.5
x_2	1	2	4	7	5	5	4.5
	$\vec{x}_1$	$\vec{x}_2$	$\vec{x}_3$	$\vec{x}_4$	$\vec{x}_5$	$\vec{x}_6$	$\vec{x}_7$

Let's $\vec{v}_1$

$k=2$: Initial states $\vec{v}_1 = (1.5, 2)$

$$\vec{v}_1 = (1, 1)$$

$$\vec{v}_2 = (3, 4)$$



Iteration 0 : (1,1) (3,4)

data points	$(1,1)$ distance from $\vec{v}_1$	$(3,4)$ distance from $\vec{v}_2$	Minimum distance	Assignment cluster center
$(1,1)$ $\vec{x}_1$	0	$\sqrt{(3-1)^2 + (4-1)^2} = \sqrt{13}$	0	$\vec{v}_1$
$(1.5, 2)$ $\vec{x}_2$	$\sqrt{5}/2 = 1.1$	$5/2 = 2.5$	1.1	$\vec{v}_1$
$(3,4)$ $\vec{x}_3$	$\sqrt{(3-1)^2 + (4-1)^2} = \sqrt{13}$	0	0	$\vec{v}_2$
$(5,7)$ $\vec{x}_4$	$2\sqrt{13}$	$\sqrt{(5-3)^2 + (7-4)^2} = \sqrt{13}$		$\vec{v}_2$
$(3,5)$ $\vec{x}_5$	$\sqrt{(3-1)^2 + (5-1)^2}$	$\sqrt{(3-3)^2 + (5-4)^2}$		$\vec{v}_2$
$(4,6)$ $\vec{x}_6$	$\sqrt{(4-1)^2 + (6-1)^2}$	$\sqrt{(4-3)^2 + (6-4)^2}$		$\vec{v}_2$
$(3.5, 4.5)$ $\vec{x}_7$	$\sqrt{(3.5-1)^2 + (4.5-1)^2}$	$\sqrt{(3.5-3)^2 + (4.5-4)^2}$		$\vec{v}_2$

$$C_1 = \{ \vec{x}_1, \vec{x}_2 \}$$

$$C_2 = \{ \vec{x}_3, \vec{x}_4, \vec{x}_5, \vec{x}_6, \vec{x}_7 \}$$

Recalculate cluster centre.

cluster centre of C_1

$$= \vec{v}_1$$

$$= \frac{1}{2} (1 + 1.5, 1 + 2)$$

$$= \underline{(1.25, 1.5)}$$

cluster centre of C_2

$$= \vec{v}_2$$

$$= \frac{1}{5} (3 + 5 + 3 + 4 + 3.5, 4 + 7 + 5 + 5 + 4.5)$$

$$= \underline{(3.7, 5.5)}$$

* Here the new cluster centers

$$\vec{v}_1$$

$$(1.25, 1.5)$$

$$\vec{v}_2$$

$$(3.7, 5.5)$$

is not same as the old cluster centers

$$v_1 = (1, 1)$$

$$v_2 = (3, 4)$$

So,

we will again iterate.

* Iteration - 1

		(1.25, 1.5)	(3.7, 5.5)		
data point		distance v_1	distance v_2	Minimum distance	Assigned cluster
(1,1)	x_1	4.5	5.24	4.5	$\vec{v}_1$
(1.5, 2)	x_2	0.5			
(3, 4)	x_3				
(5, 7)	x_4				
(3, 5)	x_5				
(4, 5)	x_6				
(3.5, 4.5)	x_7				

$C_1 =$

$C_2 =$

Recalculate the cluster centers of C_1 & C_2 .

$\vec{v}_1 =$

$\vec{v}_1 =$

$\vec{v}_2 =$

$\vec{v}_2 =$

Problem

x_1	2	1.5	3	5	3	4	3.5
x_2	1	2	4	4	5	5	4.5
x_3	2	2	2	2	2	2	2
	$\bar{x}_1$	$\bar{x}_2$	$\bar{x}_3$	$\bar{x}_4$	$\bar{x}_5$	$\bar{x}_6$	$\bar{x}_7$

$K=2$.

Perform k-means clustering with $K=2$.

Perform 2 iterations.

Iteration 0 : $(1, 1, 2)$ $(1.5, 2, 2)$

$$\sqrt{(1-5)^2 + (2-1)^2 + (2-2)^2} = \sqrt{5/2}$$

data points	$(1, 1, 2)$ distance V_1	$(1.5, 2, 2)$ distance V_2	minimum distance	Assigned clusters
$(1, 1, 2) \bar{x}_1$	0	$\sqrt{5/2}$	0	$\bar{V}_1$
$(1.5, 2, 2) \bar{x}_2$	$\frac{\sqrt{5}}{2}$	0	0	$\bar{V}_2$
$(3, 4, 2) \bar{x}_3$	$\sqrt{3} = 3.60$	2.5	2.5	$\bar{V}_3$
$(5, 7, 2) \bar{x}_4$	$2\sqrt{3} = 7.2$	6.10	6.10	$(\bar{V}_4)$
$(3, 5, 2) \bar{x}_5$	$2\sqrt{5} = 4.47$	3.3	3.3	$\bar{V}_5$
$(4, 5, 2) \bar{x}_6$	5	3.9	3.9	$\bar{V}_6$
$(3.5, 4.5, 2) \bar{x}_7$	4.3	3.2	3.2	$\bar{V}_7$

Hierarchical clustering

Two types

- └ Bottom up
- └ Top down

① Bottom up

It builds a dendrogram (Tree) from the bottom level to the top level (bottom-up approach)

~~In this approach most similar~~

Algorithm/method

1. Initially each data point forms a cluster.
2. Compute the distance b/w clusters.
3. Repeat until a single cluster remains

{ Merge the two closest clusters.

Update distance matrix.

}

different definitions to the distance leads to different algorithms

① Single linkage clustering

Smallest distance between an element in one cluster and element in another cluster

② Complete linkage Clustering

maximum distance .

③ Average linkage.

(a) PROBLEM

Consider the below distance table

	A	B	C	D
A	0	10	12	20
B	10	0	5	3
C	12	5	0	4
D	20	3	4	0

Perform hierarchical clustering [single linkage] and show the dendrogram.

	A	B	C	D
A	0	10	12	20
B	10	0	5	3
C	12	5	0	4
D	20	3	4	0

min. distance $\Rightarrow 3$.

merge B & D

{BUD}

	A	{BUD}	C
A	0	10	12
{BUD}	10	0	4
C	12	4	0

Recalculate the distance.

diagonal $\Rightarrow 0$

A to {BUD}

$$d(\{A\} \{BUD\}) = \min \{d(A, B) \cup d(A, D)\}$$

$$= \min \{10, 20\}$$

$$= 10 //$$

$$d(\{C\} \{BUD\}) = \min \{d(C, B) \cup d(C, D)\}$$

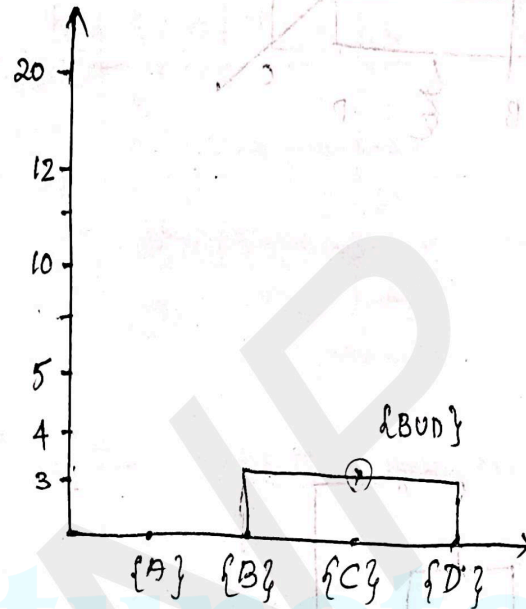
$$= \min \{5, 4\}$$

$$= 4 //$$

~~update~~ Iteration 0

	A	{BUD}	C
A	0	10	12
{BUD}	10	0	(4)
C	12	(4)	0

min (4)



Iteration 1

	A	{BUD}	C
A	0	10	12
{BUD}	10	0	(4)
C	12	(4)	0

$$d(\{C\} \cup \{BUD\}) = \min\{d(C, BUD),$$

$$d(C,$$

$$d(\{C\} \cup \{BUD\}) = \min\{d(A, \{BUD\}) \cup$$

$$d(d(A) \{BUD\})$$

$$= \min\{10, 12\}$$

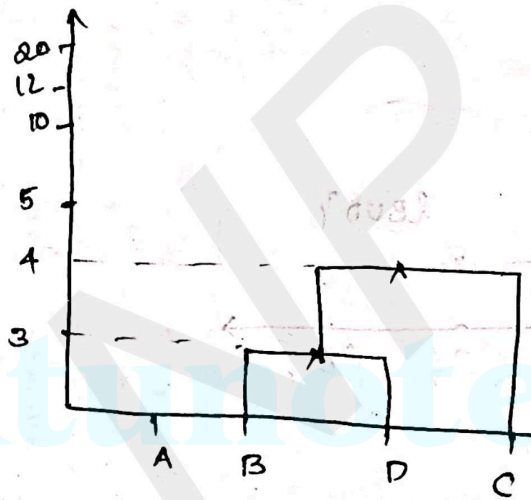
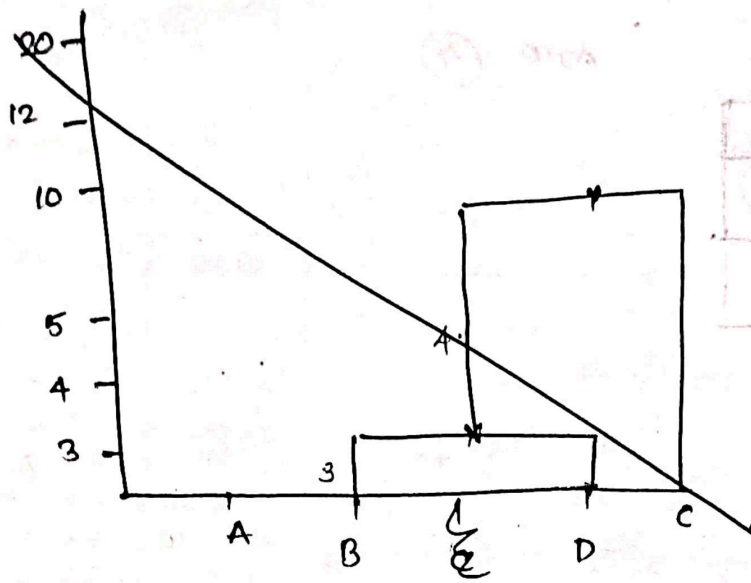
$$= 10 //$$

A {BUD} U {C}

A	{BUD} U {C}
0	10
{BUD} U {C}	10 0

{BUD} U {C}

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9/11/2022

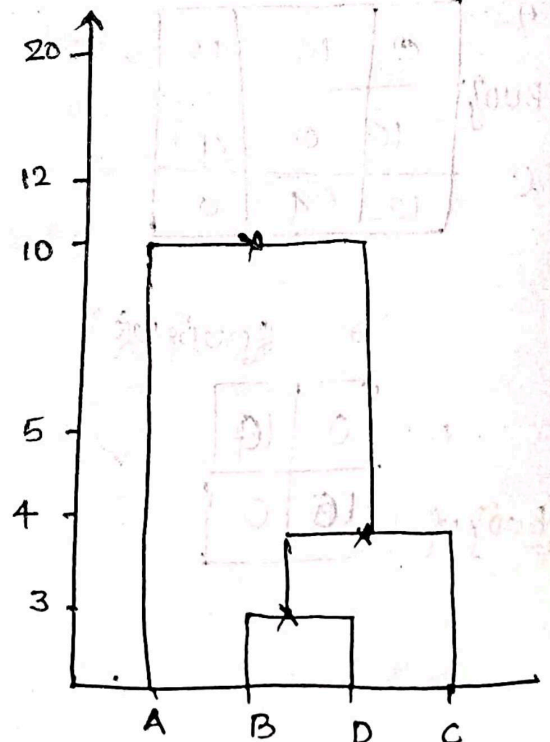


Iteration 2

{Budget}

	A	
A	0	6
	6	0

{Budget}



Complete Linkage.

Given a distance matrix.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

construct a dendrogram using single linkage clustering

Iteration 0

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

merge $\{c\} \cup \{e\}$

update -

	a	b	cue	d
a	0	9	3	6
b	9	0	6	5
cue	3	6	0	8
d	6	5	8	0

$$d(\{a\} \cup \{c, e\}) =$$

$$\min \{d(a, c), d(a, e)\}$$

$$\min \{3, 11\} = 3 //$$

$$d(\{b\} \cup \{c, e\})$$

$$\min \{d(b, c), d(b, e)\}$$

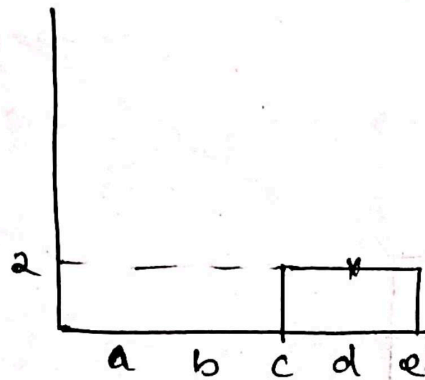
$$\min \{7, 10\} = 6 //$$

$$d(\{d\} \cup \{c, e\})$$

$$\min \{d(d, c), d(d, e)\}$$

$$\min \{9, 8\} = 8 //$$

dendrogram



Iteration - 1

$d(a, \{b, d\})$

		0	8	6
b		8	0	5
d		6	5	0

min $\rightarrow 3$

$$\begin{aligned} d(a, \{b, d\}) &= \\ \min(d(a, b), d(a, d)) \end{aligned}$$

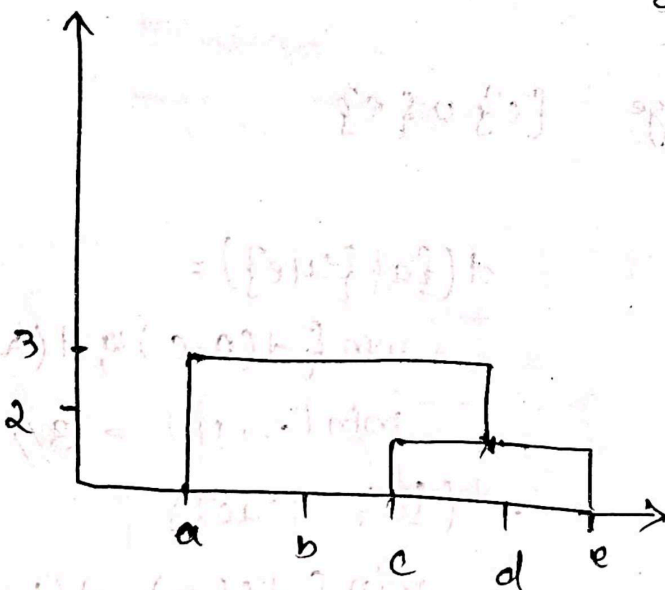
$$d(a, \{b, d\})$$

$$= \min(d(a, b), d(a, d))$$

$$= \min(8, 6)$$

$$= 6 //$$

dendrogram



$$d(b, \{c, d\})$$

$$= \min(d(b, c), d(b, d))$$

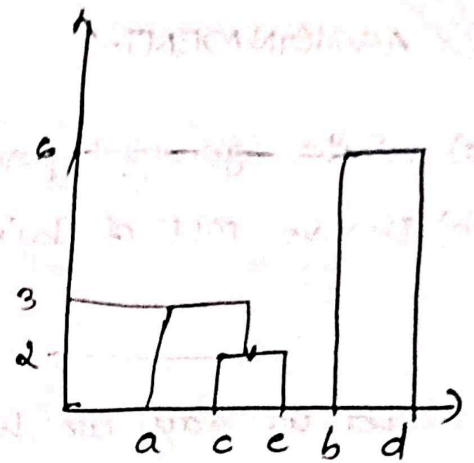
$$= \min(9, 8)$$

$$= 8 //$$

Iteration - 2.

min 5

	$\{a \cup \{cue\}\}$	bud
av(cue)	0	6
bud	6	0



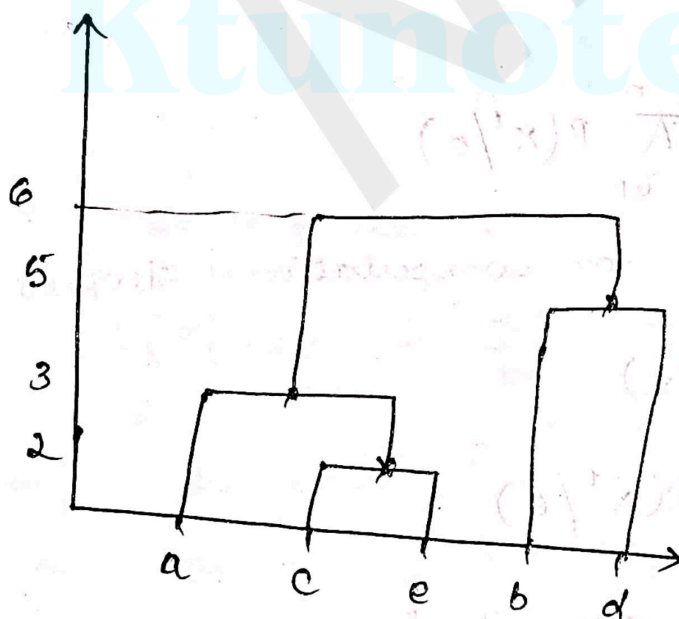
$$d(av(cue), bud) =$$

$$= \min \{ d(av(cue), b), d(av(cue), d) \}$$

$$= \min (3, 6)$$

$$= \underline{\underline{3}}$$

dendrogram



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Construct dendrogram using the below matrices.
use complete linkage.

	a	b	c	d	e
a	0	-	-	-	-
b	8	0	-	-	-
c	2	6	0	-	-
d	5	4	8	0	-
e	10	9	1	7	-

Iteration - 0

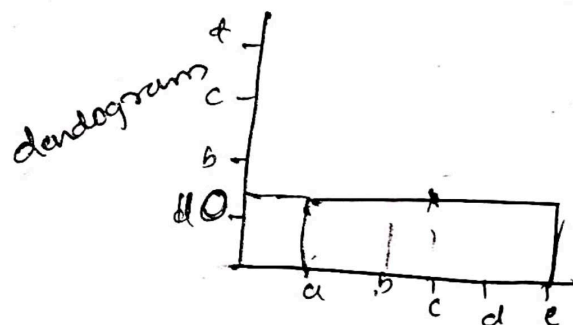
	a	b	c	d	e
a	0	-	-	-	-
b	8	0	-	-	-
c	2	6	0	-	-
d	5	4	8	0	-
e	10	9	1	7	-

update

	$\{e\} \cup \{a\}$	b	c	d
$\{e\} \cup \{a\}$	0	9	2	7
b	9	0	6	4
c	2	6	0	8
d	7	4	8	0

merge.

$\{e\} \cup \{a\}$.



$$d(\{e\} \cup \{a\}, b)$$

$$= \max \{ d(\{e\}, \{b\}), d(\{a\}, \{b\}) \}$$

$$= \max(9, 8)$$

$$= 9 //$$

$$d(\{e\} \cup \{a\}, c) = \max(1, 2) = 2 //$$

$$d(\{e\} \cup \{a\}, d) = \max(7, 5) = 7 //$$

Iteration - 1

$(\{e\} \cup \{a\}) \cup b$ c d

$\{e\} \cup \{a\}$			
c	0	6	7
d	6	0	8
	7	8	0

$$d(\{e\} \cup \{a\} \cup b, c)$$

$$= \max(d(\{e\} \cup \{a\}, b), d(\{e\} \cup \{a\}, c))$$

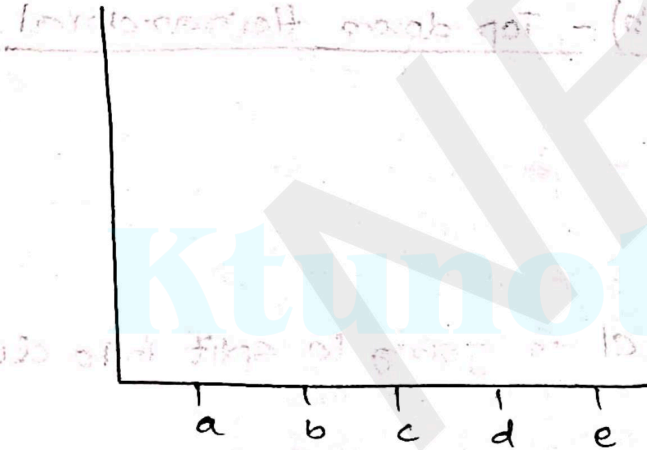
$$= \max(2, 6)$$

$$= 6 //$$

$$d(\{e\} \cup \{a\} \cup b, d)$$

$$= \max(7, 4) = 7 //$$

dendrogram



Iteration - 2

$(\{e\} \cup \{a\}) \cup b$ $c \cup d$

$\{e\} \cup \{a\}$		
$c \cup d$	0	7
	7	0

$$\max(d(c \cup d, (\{e\} \cup \{a\}) \cup b))$$

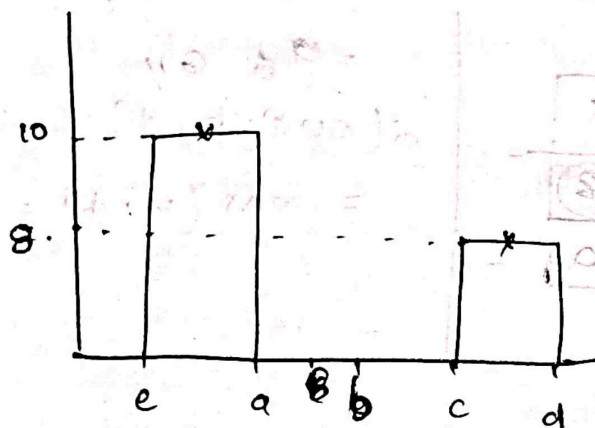
$$= \max(c, (\{e\} \cup \{a\}) \cup b),$$

$$d(\{e\} \cup \{a\} \cup b)$$

$$= \max(6, 7)$$

$$= 7 //$$

dendrogram



16/11/2022

DIANA (Divisive Analysis) - Top down Hierarchical clustering

Algorithm

Step 1: Suppose that cluster C_1 is going to split into clusters C_i & C_j

Step 2: Let $C_i = C_1$, $C_j = \phi$

Step 3: For each object $x \in C_i$:

(a) In the 1st iteration compute average distance of x to all other objects.

Move the object with max Average distance to C_j

(b) For the remaining iterations
Compute $D_x = \text{average} \{d(x, y), y \in C_i\} - \text{average} \{d(x, y), y \in C_j\}$



~~Step 4~~: Find an object x in C_i for which D_x is largest.

If $D_x > 0$ then move x to C_j

Step 4: Repeat step (3)b until all D_x are -ve. Then C_i is

~~Step 5~~: split into C_i & C_j

Step 5: Select the smaller clusters with largest diameter. Then, repeat the step ① → ④

Problem

Q) Given a data set $\{a, b, c, d\}$ with the distance matrix

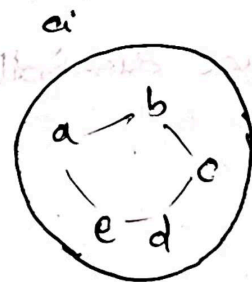
	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

1) $C_1 = \{a, b, c, d, e\}$

2) $C_i = C_1$

$C_j = \phi$

$= \{a, b, c, d, e\}$



3) Initial iteration

calculate the average dissimilarity of objects c_i with other objects in C_i using the average dissimilarity measured.

Average dissimilarity of $a = \frac{1}{4} \{d(a,b) + d(a,c) + d(a,d) + d(a,e)\}$

$$= \frac{1}{4} \{9 + 3 + 6 + 11\}$$

$$= \frac{1}{4} \{29\}$$

$$= \underline{\underline{7.25}}$$

Average dissimilarity of $b = \frac{1}{4} \{d(b,a), d(b,c), d(b,d), d(b,e)\}$

$$= \frac{1}{4} \{$$

$$= \underline{\underline{7.75}}$$

Average dissimilarity of $c = \frac{1}{4} \{d(c,a), d(c,b), d(c,d), d(c,e)\}$

$$= \frac{1}{4} \{9 + 7 + 9 + 2\}$$

$$= \underline{\underline{5.25}}$$

Average dissimilarity of $d = \frac{1}{4} \{d(d,a), d(d,b), d(d,c), d(d,e)\}$

$$= \frac{1}{4} \{6 + 5 + 9 + 8\}$$

$$= \underline{\underline{7}}$$

Average dissimilarity of $e = \frac{1}{4} \{d(e,a), d(e,b), d(e,c), d(e,d)\}$

$$= \frac{1}{4} \{11 + 10 + 2 + 8\}$$

$$= 7.75$$

Here, highest Avg distance is 7.75. for objects b and we choose one of them

Let,

It is b

then,

we move {b} to C_i

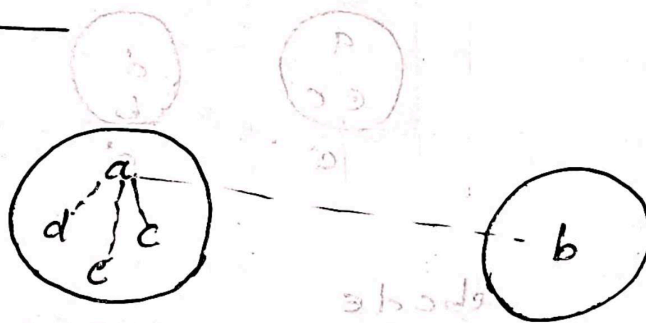
So,

$$C_i = \{a, c, d, e\}$$

$$C_j = \{b\}$$

For the remaining iteration.

Iteration 1



$$D_a = \frac{1}{3} (d(a,d) + d(a,e) + d(a,c)) - \frac{1}{4} \{d(a,b)\}$$

$$= \frac{20}{3} - 9$$

$$= -2.33$$

$$D_d = \frac{1}{3} (d(d,a) + d(d,c) + d(d,e)) - \frac{1}{4} \{d(d,b)\}$$

=

$$= \underline{\underline{0.67}} \checkmark$$

$$D_c = \frac{1}{3}(d(c,a) + d(c,d) + d(c,e)) - \frac{1}{3}d(e,b)$$

=

$$= \underline{\underline{-0.33}}$$

$$D_e = \frac{1}{3}(d(e,a) + d(e,c) + d(e,d)) + \frac{1}{3}d(e,b)$$

$$= \underline{\underline{0}}$$



$$c_i = \{a, c, e\}$$

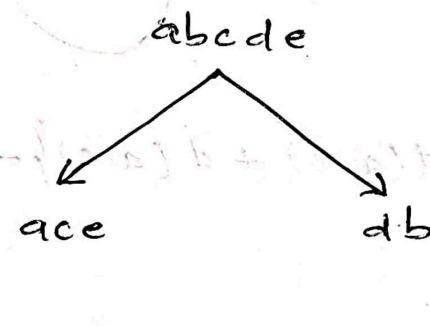
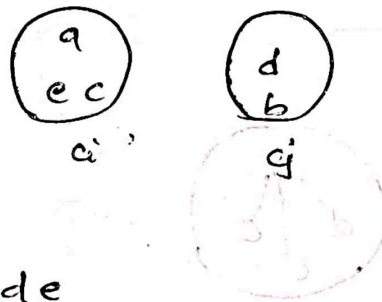
$$c_j = \{b, d\}$$

Iteration - 2

$$D_a = -0.5$$

$$D_e = -13.5$$

$$D_c = -2.5$$



Diameter of c_i is larger than c_j , so split c_i

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$$\begin{aligned}\text{Diameter of } C_i &= \max \{d(a,c), d(a,e), d(c,e)\} \\ &= \max \{3, 11, 2\} \\ &= \underline{\underline{11}}\end{aligned}$$

$$\begin{aligned}\text{Diameter of } C_j &= \max \{d(b,d)\} \\ &= \underline{\underline{5}}\end{aligned}$$

Then we will divide $C_i = \{a, c, e\}$

Dimensionality Reduction

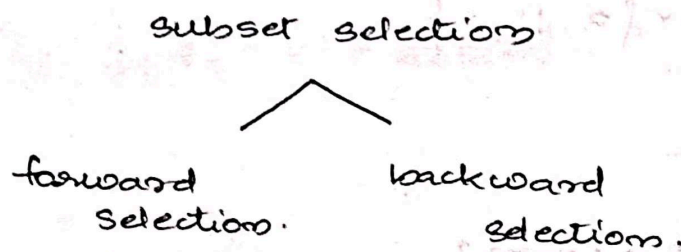
- * The complexity of any classifier or regressor depends on the number of input features (dimension)
 - * Dimension determines both time and space complexity of the algorithm (training algorithm)
 - * By reducing the dimension the performance of the classifier can be improved. This is called dimensionality reduction
- two methods.

1. Feature selection
2. Feature extraction.

1. Feature selection

In the feature selection we are interested in finding k of the d dimensions that gives the most information and we discard other $d - k$ dimension.

Subset selection method is commonly used for feature selection



2. Feature Extraction

In the feature extraction, we are interested in finding a new set of k-dimensions that are a combination of d dimensions.

eg:- d: $[d_1 \ d_2 \ d_3]$

$2 = k \rightarrow (d_1, d_2)$
 (d_2, d_3)

Methods may be supervised or unsupervised.

- PCA [Principal Component Analysis]

↖ unsupervised method.

- LDA [Linear Discriminant Analysis]

↖ supervised method

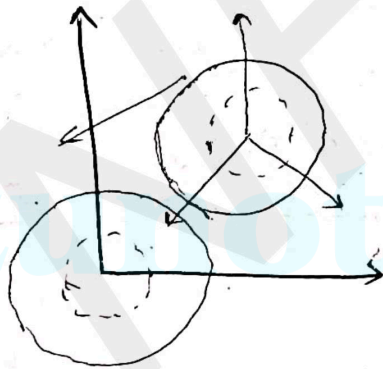
- Isomap [Isometric feature mapping]

- LLE [Local linear embedding.]

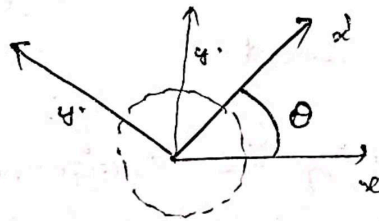
PCA [Principal Component Analysis]

- * It is an unsupervised method used for dimensionality reduction through feature extraction method.
- * It uses the concept of projection.

Given a set of samples. the PCA algorithm ^{Centers} sends the samples.

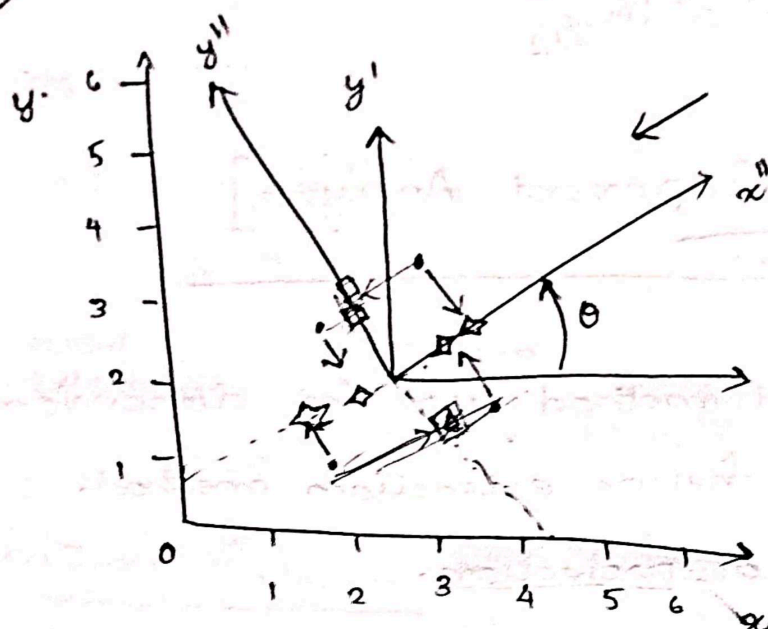


Rotate the axis at an angle θ



Such that the projection of samples on the new axis should show maximum variance

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PCA.

1. Centre.
2. Rotate θ .

- projection on y axis y'
- ◇ projection on x axis x'

Let,

Z_1 and Z_2 are the new axis formed

If variance of data on Z_2 is small compared to variance on Z_1 . Then Z_2 can be ignored.

STEP Centre the axis and

1. Create new axis by rotating the original axis at an angle θ such that new axis are [Principal components] are orthogonal to each other.

2. Then project the data points on to new axis using the transformation

$$Z = W^T X$$

3. Calculate the variance of data on all axes (Principal

components]

Axis with greatest variance is taken as first principal component and so on.

$$\lambda(Z_1) > \lambda(Z_2) > \dots$$

Eigen vector

Expectation Maximization (EM)

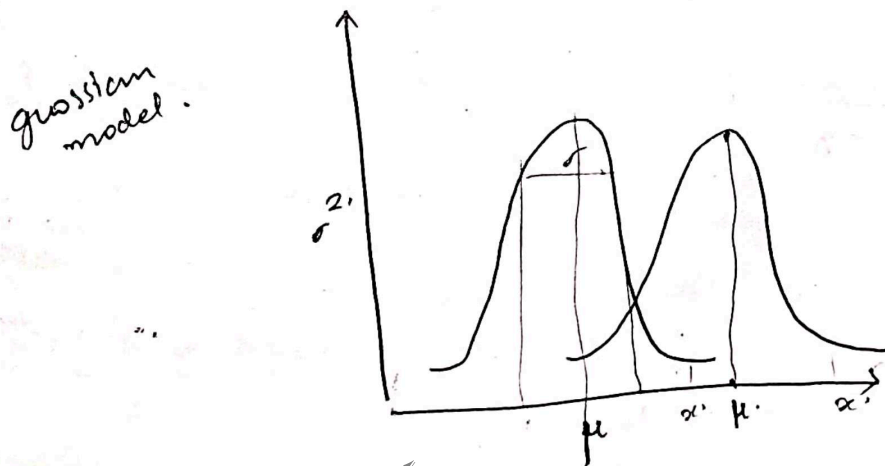
- * It uses the concept of maximum likelihood estimation
- * It can be used for clustering
- * It uses probabilistic models
[Normal density function / Gaussian density function etc]
- * It is an Gaussi Iterative algorithm.
- * It consists of two steps

E-step and (Expectation step)

M-step (Maximization step)

E-step

Initialize the clusters C_i & C_j and choose a probabilistic model (Gaussian etc) and initialize parameters.



eg:- μ, σ^2, σ
 $N \& \sigma$

Perform E-step and M-step repeatedly.

E-step:

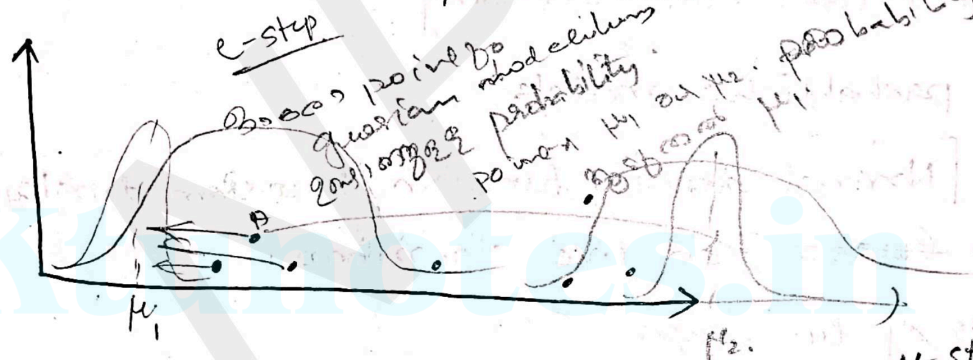
For each data point calculate the probability that the datapoint belongs to cluster C_i and C_j

M-step

Assign datapoint to the nearest cluster that is cluster with maximum probability.

Recalculate the cluster parameters using the assigned points and try to maximize these parameters.

eg:-



Suppose we need to form 2 clusters
Choose probabilistic model (Gaussian)
Initialize 2 Gaussian models.

μ_1, σ^2 with (μ_1, σ_1) & (μ_2, σ_2)

E-step.

M-step
Points on x-axis
assign μ_1 & μ_2
each assign points
then avg points
avg avg points
we calculate new
we move on x-axis based
on points

$$P(x) = \underline{N(\mu, \sigma^2)} = \frac{1}{\sqrt{\sigma^2 2\pi}} e^{-\dots}$$